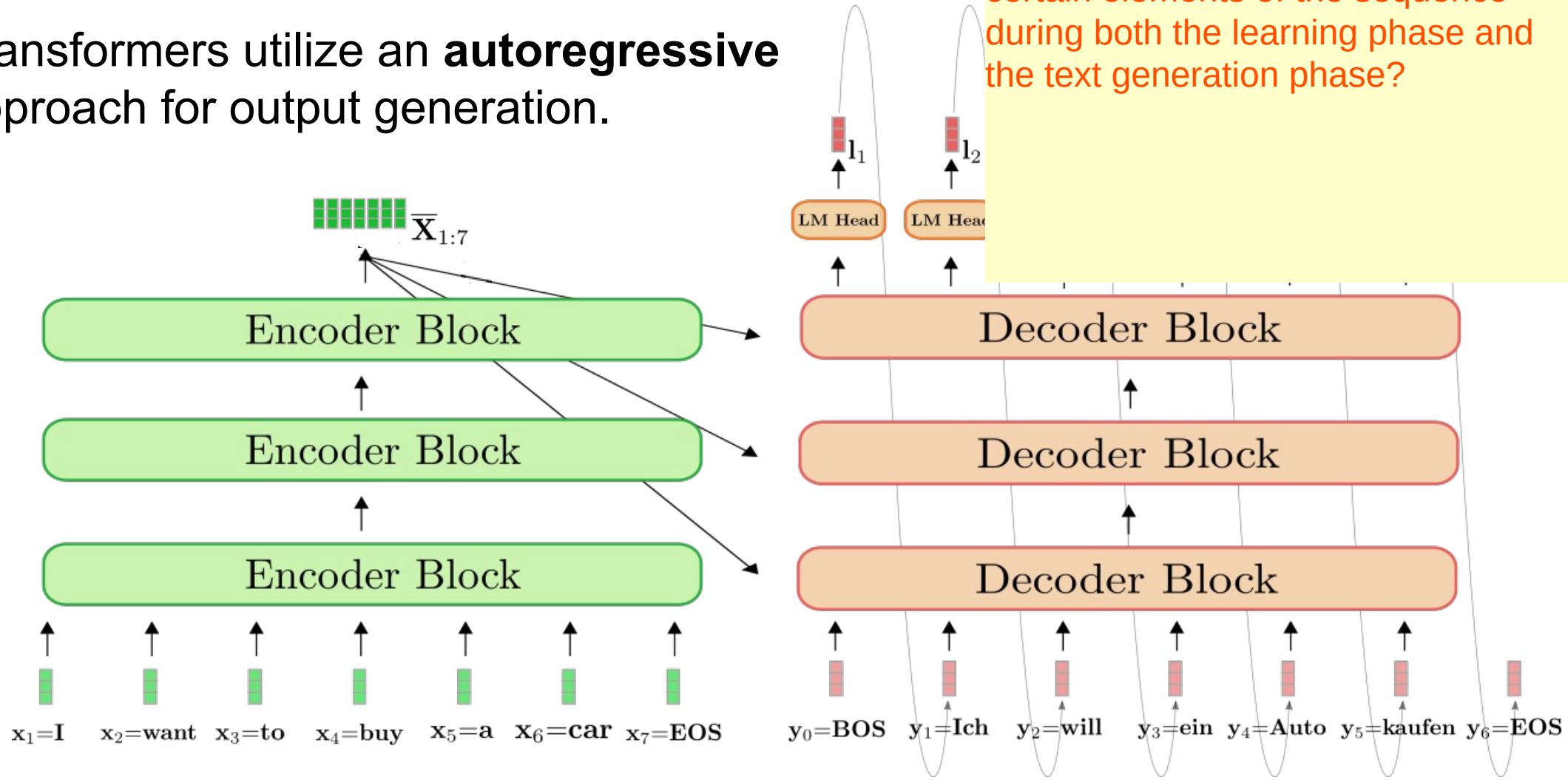
Example:
Language Translation with Trans

QUESTION: Why do we intentionally block the architecture from evaluating certain elements of the sequence during both the learning phase and the text generation phase?

Transformers utilize an **autoregressive** approach for output generation.

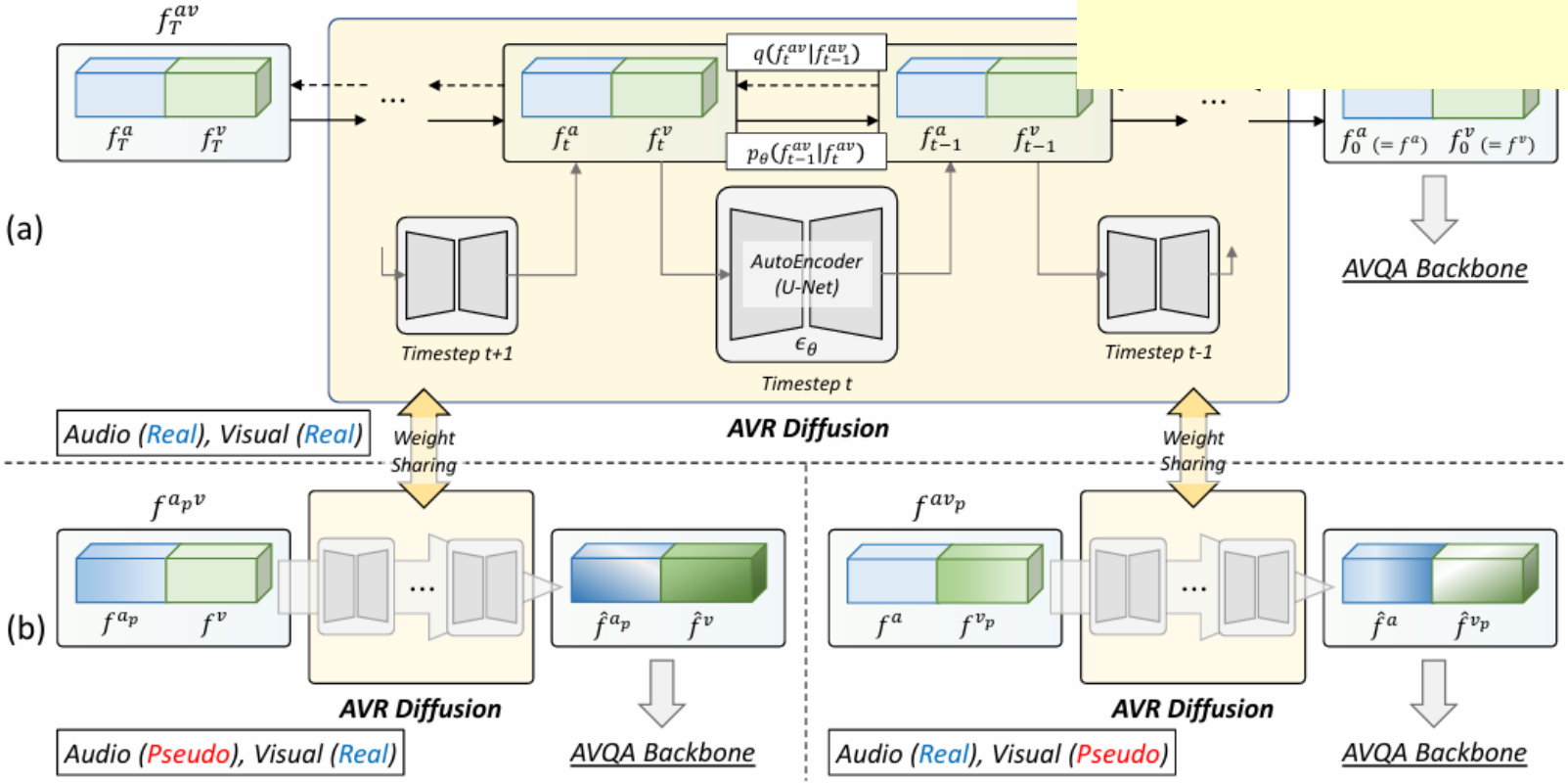


Source: https://huggingface.co/blog/encoder-decoder

Method 1: Modality Imputation

QUESTION: When attempting to restore incomplete audio-visual streams, how is the backward noise-reduction step visually differentiated when it combines authentic input with artificially generated substitute data?

- AVR: Audio-Visual Relation-aware diffusion m
  - (a) depicts the diffusion and reverse process of concater features, while (b) represents the reverse process of feat and the real feature are concatenated.



mRAG
(Multimodal Retrieval Augmente

QUESTION: Based on a cited 2022 NAACL paper, what are the three distinct streams of information required to successfully predict an answer when combining visual comprehension with outside databases?

- mRAG aims to provide MLLMs with more co
  and up-to-date world knowledge from exter
- Two categories of mRAG based on the retrie
  - (1) One category employs visual encoding mode
    representation, and then retrieves the knowledge from knowledge base with the highest feature similarity.

Dynamic questions

QUESTION: What specific flaw is introduced when a system clumsily merges the user's prompt with the written descriptions of visual elements to form a single, giant search string?

- Dynamic questions are defined as questions t flexibly provide knowledge retrieval solutions and time of retrievals are all variable.
  - These questions cannot be solved by a predefined retrieval process.
- Example types of dynamic questions:
  - (1) Questions with rapidly changing answers
  - (2) Questions requiring multi-modal knowledge
  - (3) Multi-hop questions

The prompt used in original MLLMs, heuristic mRAG-empowered MLLMs, and the OmniSe

QUESTION: In the introduced benchmark collection for information-seeking tasks, what exact temporal metric is used as the threshold to categorize a fact's evolution speed as either sluggish or rapid?

Model	Lang.	Prompt
MLLMs	en	Q{input_image} {input_question} Answer:
	zh	Q{input_image} {input_question} 答案:
MLLMs + Heuristic mRAGs	en	Below are related image-text pairs searched, which may be helpful for answering questions later on: {retrieved_images}{retrieved_texts} Refer to the above background knowledge and answer the following question: Q {input_image} {input_question} Answer:
	zh	以下是检索到的相关图文对，可能对后面回答问题有帮助: {retrieved_images}{retrieved_texts} 参考上述背景知识，回答下面的问题: Q {input_image} {input_question} 回答: